THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL UNIT TEST SET 2

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A path 2 m wide runs around the outside of a 20 m by 15 m rectangular garden. The area of the path is: 1

(a) 156 m² **(b)** 456 m² **(c)** 140 m² **(d)** 16 m²

2. A cube of side 4 cm is painted all over and cut into 1 cm cubes. The number of cubes with no face painted is: 1

(a) 56 **(b)** 4 **(c)** 24 **(d)** 8

3. The mean of the first 9 multiples of 4 (that is 4, 8, ..., 36) is: 1

(a) 20 **(b)** 18 **(c)** 22 **(d)** 40

4. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 53 **(b)** 109 **(c)** 67 **(d)** 81

5. A right triangle has hypotenuse 25 cm and one leg 15 cm. Its area is: 1

(a) 300 cm² **(b)** 150 cm² **(c)** 187 cm² **(d)** 250 cm²

6. A Koch curve starts with a line of length 27 cm. The total length after stage 2 (in cm) is: 1

(a) 72 **(b)** 54 **(c)** 48 **(d)** 15.19

7. On a line graph, a town's population (in thousands) rises from 20 to 30. The percentage increase is: 1

(a) 25% **(b)** 60% **(c)** 50% **(d)** 10%

8. The sum of 3 consecutive even numbers starting with $2n$ is: 1

(a) $6n + 6$ **(b)** $6n + 3$ **(c)** $3n + 6$ **(d)** $6n + 8$

9. The diagonal of a square is 10 cm. Its area is: 1

(a) 25 cm² **(b)** 100 cm² **(c)** 20 cm² **(d)** 50 cm²

10. The total number of triangles removed from the Sierpinski triangle up to stage 3 is: 1

(a) 26 **(b)** 9 **(c)** 27 **(d)** 13

11. The mean mark of 20 students is 35 and of another 30 students is 45. The mean of all 50 students is: 1

(a) 2050 **(b)** 41 **(c)** 40 **(d)** 43

12. On expanding $5(x + 2) + 4x$ we get: 1

(a) $9x + 10$ **(b)** $5x + 14$ **(c)** $9x + 2$ **(d)** $20x + 10$

13. An L-shaped figure is made by removing a 10 cm by 7 cm rectangle from a 25 cm by 18 cm rectangle. Its area is: 1

(a) 520 cm² **(b)** 380 cm² **(c)** 450 cm² **(d)** 165 cm²

14. A prism has 24 edges. The number of its faces is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

15. The mean of 20, 39, 32, 34, 52, 27 is 34. If 32 is replaced by 44, the new mean is: 1

(a) 36 **(b)** 41.33 **(c)** 46 **(d)** 34

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The mean of 5 observations 27, 38, 9, 33, 28 is 27. Find the mean if 6 is added to every observation. 2

17. Expand $(x + 4)(x + 9)$. 2

18. A rhombus has side 50 cm and one diagonal 28 cm. Find its other diagonal and area. 2

19. A Koch curve starts with a line of length 63 cm. Find the number of segments and the total length at stage 2. 2

20. Find the mean of the first 10 natural numbers that are multiples of 3. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A parallelogram has base 24 cm and height 10 cm. Another side of 16 cm has a corresponding height; find that height and the area. 3

22. A Sierpinski triangle starts with a triangle of area 128 cm². Find the shaded area and the number of shaded triangles at stage 3. 3

23. The median of 60, 47, 76, 68, 36, 49, 21 is 49. If the largest value 76 is changed to 228, what happens to the median and the mean? 3

24. Show that the sum of 3 consecutive even numbers starting with $2n$ is $6n + 6$ and verify it for $n = 4$. 3

OR

A right triangle has legs 9 cm and 12 cm. Find its hypotenuse and area, and the length of the altitude drawn to the hypotenuse. 3

25. A cube of side 5 cm is painted all over and cut into 1 cm cubes. Find the number of cubes with 3

(i) exactly three painted faces
(ii) no painted face
(iii) at least one painted face.

OR

The mean weight of 20 students is 36 kg and of another 30 students is 41 kg. Find the mean weight of all of them. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A Koch snowflake is built from an equilateral triangle of side 9 cm. Find its perimeter at stage 0, stage 1 and stage 2. **5**

OR

A class of 20 students has a mean mark of 50. Another 30 students with mean 60 join it. Find the mean of the combined group and the change in the original mean. **5**

27. Think of a two-digit number with digits a and b. Add its reverse. Show that the sum is 11(a+b) and find the sum for the digits 3, 9. Is the sum divisible by 11? **5**

OR

The parallel sides of a trapezium are 32 cm and 18 cm and its area is 250 cm². Find the distance between the parallel sides. A parallelogram of base 18 cm (the shorter parallel side) and the same height is cut out, leaving a triangle; find the area of the parallelogram and of the triangle left. **5**

END OF QUESTION PAPER — Total: 50 Marks

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